Seaborn EDA: Store Transactions

Week 4 Demo: Choosing a Predictor



Some graphics and datasets used in this presentation and accompanying MS Excel models are created by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Setup

Run once at the top of the notebook

Managers read 13,000, not 13000. One line per axis adds commas or dollar signs.

Command

```
import pandas as pd
import seaborn as sns

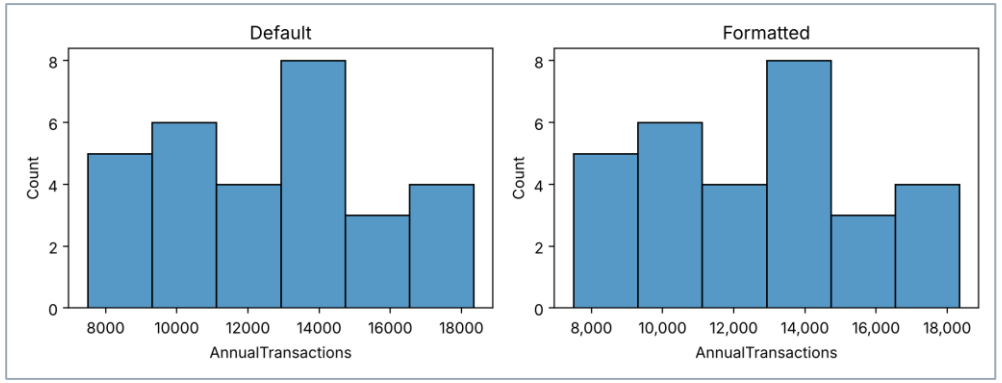
stores = pd.read_csv(
    "PP_week4_EDA_demo_stores_ChatGPT_261025.csv")
```

Format → axis shows

`"{x:,.0f}"` 13,000

`"${x:,.0f}"` $85,000

`fmt=".2f"` 0.89 (heat map cells)



Question 1

What does the distribution of annual transactions tell us about comparable stores?

Histogram

Numerical · 1 variable

Transactions run from 7,500 to 18,340, centered near 13,000. Roughly symmetric, no extreme outliers.

Command

```
ax = sns.histplot(data=stores,
        x="AnnualTransactions", kde=True)
ax.xaxis.set_major_formatter("{x:,.0f}")
```

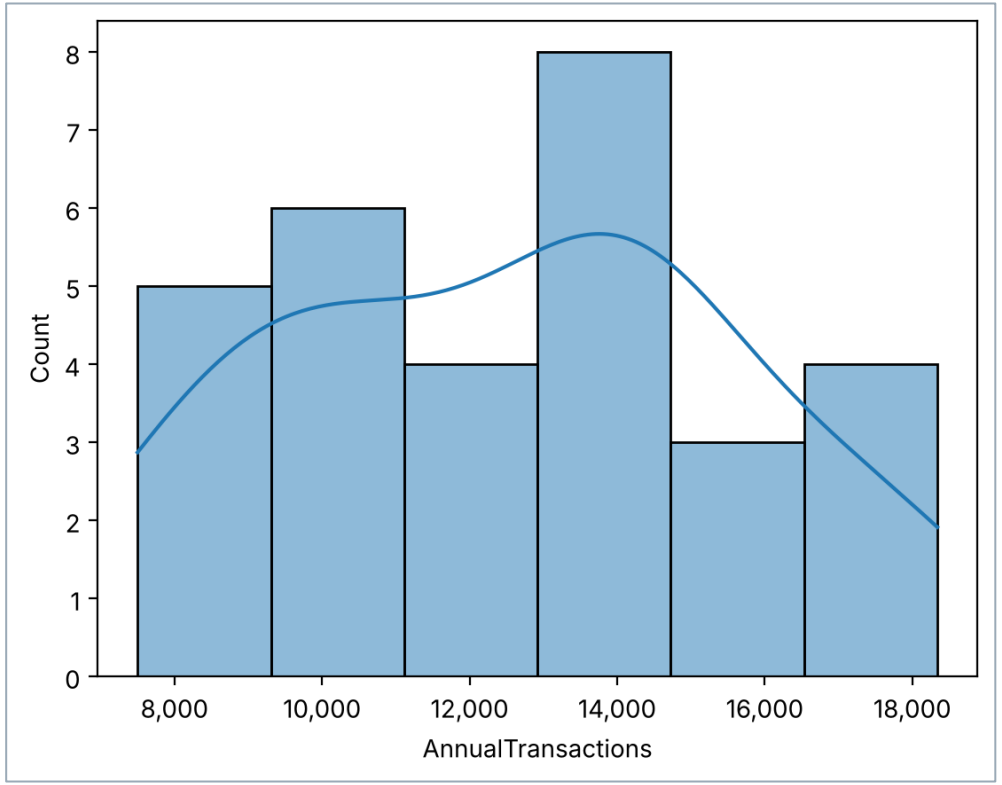
Arguments

`data=` the DataFrame

`x=` the column to plot

`kde=True` adds a smooth curve over the bars

`bins=` number of bars (optional)



Box Plot

Categorical + Numerical · 2 variables

Standard stores average about 13,500 transactions vs 11,600 for Compact, but the ranges overlap a lot.

Command

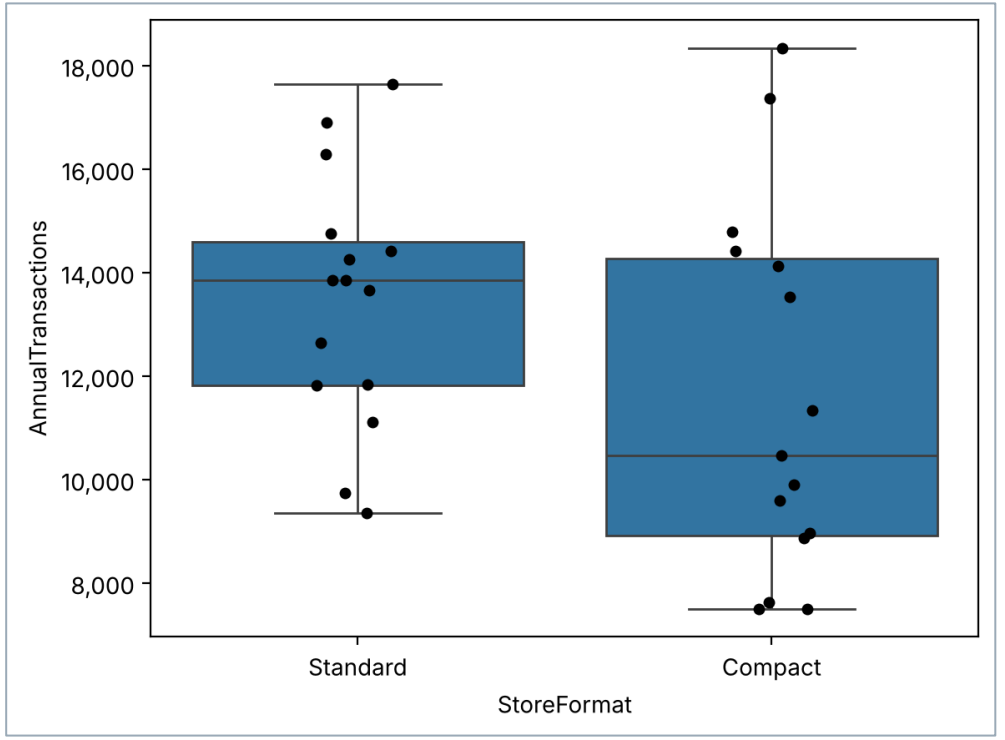
```
ax = sns.boxplot(data=stores, x="StoreFormat",
        y="AnnualTransactions")
sns.stripplot(data=stores, x="StoreFormat",
        y="AnnualTransactions", color="black")
ax.yaxis.set_major_formatter("{x:,.0f}")
```

Arguments

`x=` category, one box per group

`y=` the numeric values

`color=` dot color for the strip plot



Pair Plot (full grid)

Numerical · 4 variables, colored by 1 categorical

Every pair of variables at once, with each one's distribution on the diagonal. Read the bottom row: Trans_K vs each predictor.

Command

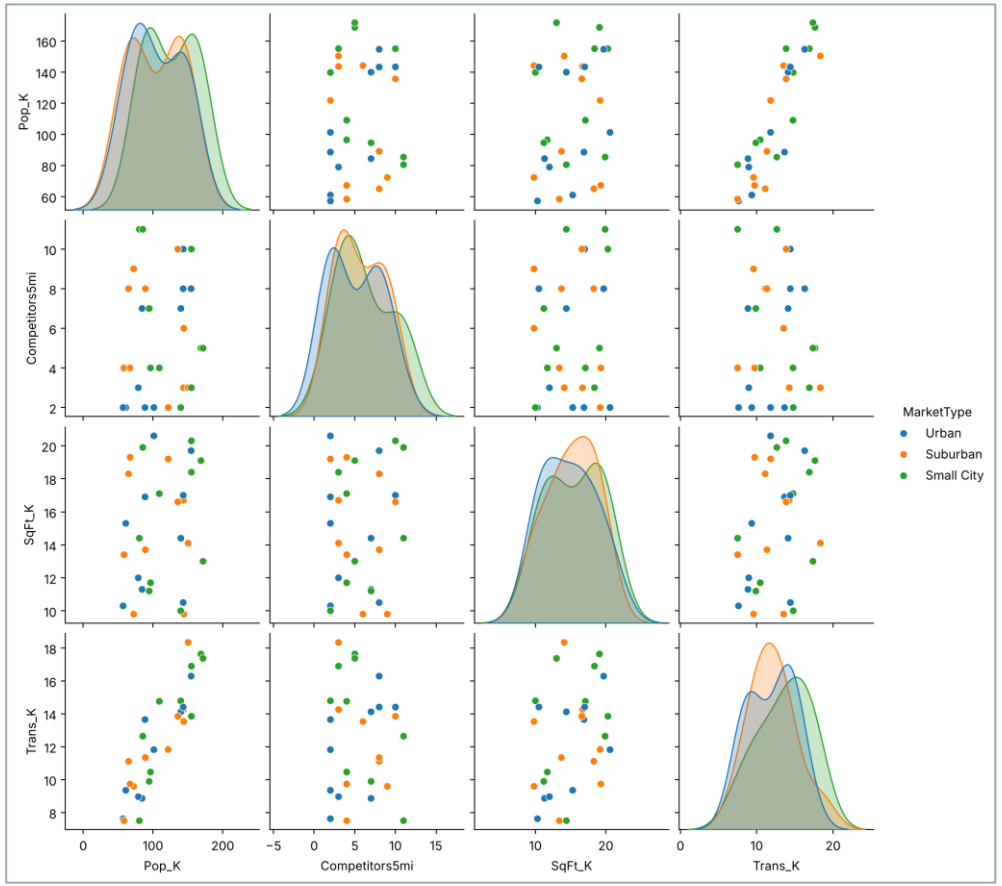
```
pp = stores.assign(
    Pop_K = stores["Population5mi"]/1000,
    SqFt_K = stores["StoreSqFt"]/1000,
    Trans_K = stores["AnnualTransactions"]/1000)
sns.pairplot(data=pp, hue="MarketType",
    vars=["Pop_K", "Competitors5mi",
          "SqFt_K", "Trans_K"])
```

Arguments

`vars=` which columns (default: all numeric)

`hue=` colors by a category

`assign()` makes K columns, simpler than formatting 16 panels



Count Plot

Categorical · 1 variable

Exactly 10 stores per market type. Balanced groups make comparisons fair, but each group is small.

Command

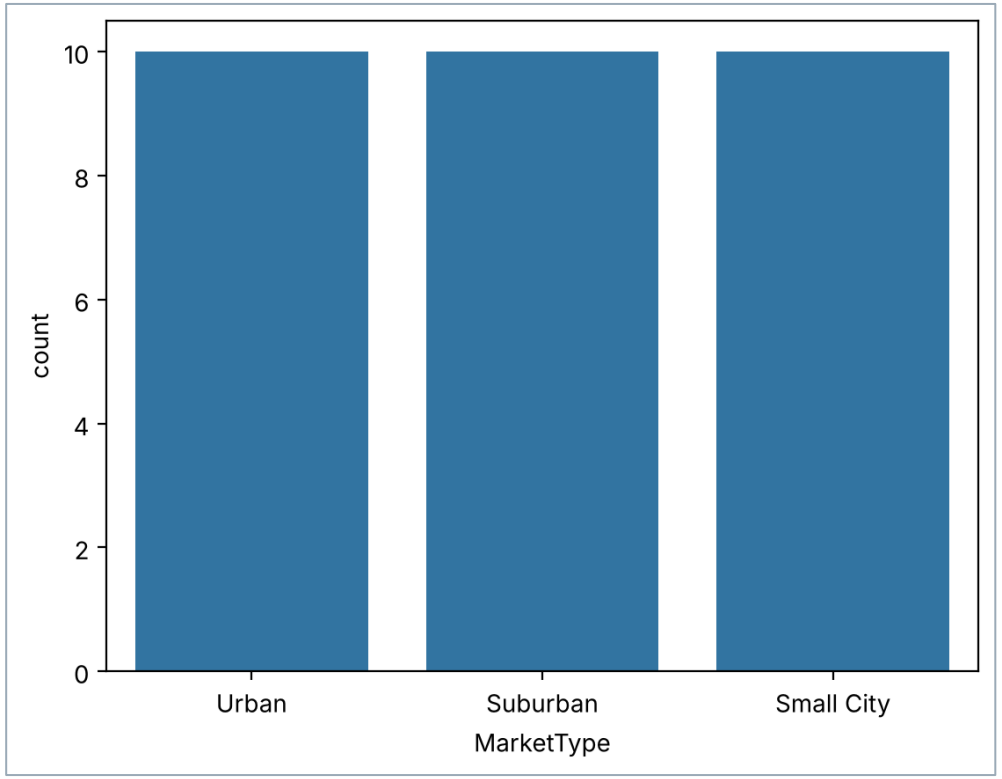
```
sns.countplot(data=stores, x="MarketType")
```

Arguments

`x=` the category to count

`(no y)` bar height is the count

`order=` list to set the bar order (optional)



Question 2

Which market characteristics appear most strongly associated with annual transactions?

Heat Map

Numerical · many variables (correlations)

Read the AnnualTransactions row.
Population5mi stands out at 0.89. Ignore AnnualSales: it is calculated from transactions.

Command

```
sns.heatmap(stores.corr(numeric_only=True),
            annot=True, fmt=".2f",
            cmap="coolwarm", vmin=-1, vmax=1)
```

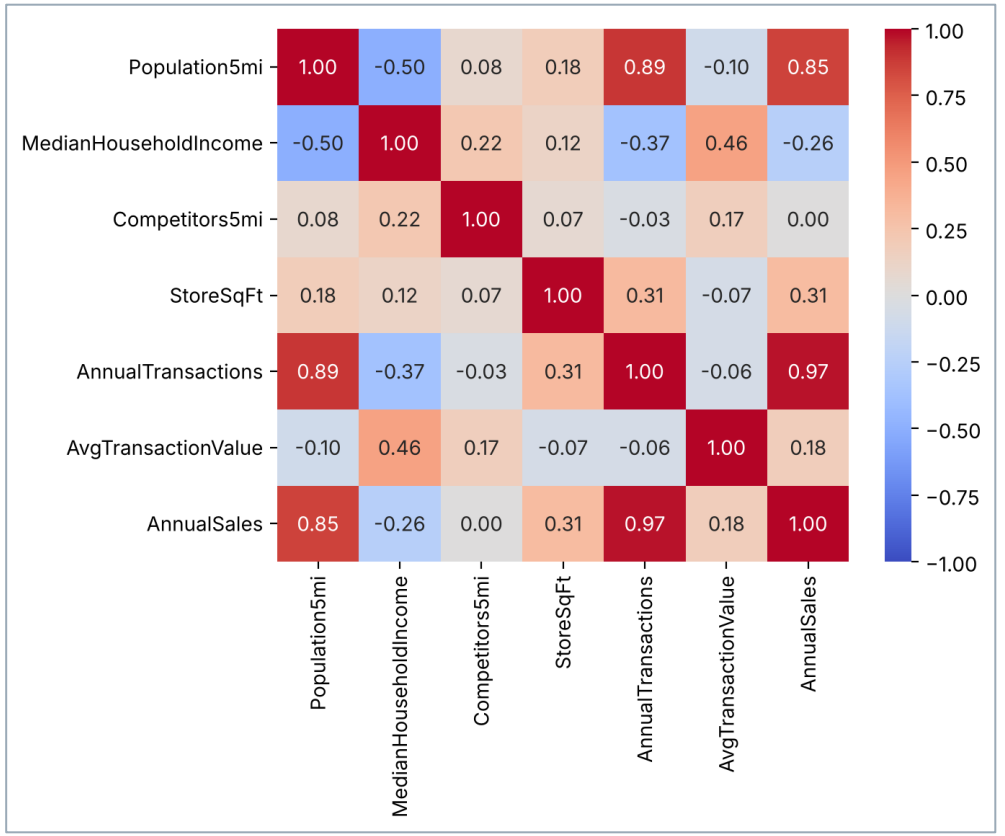
Arguments

`1st arg` the correlation table, not raw data

`annot=True` prints the numbers

`fmt=".2f"` two decimals

`vmin, vmax` fix the scale at -1 to 1



Scatter Plot

Numerical · 2 variables (color by a category)

More people within 5 miles, more transactions. The pattern holds across all three market types.

Command

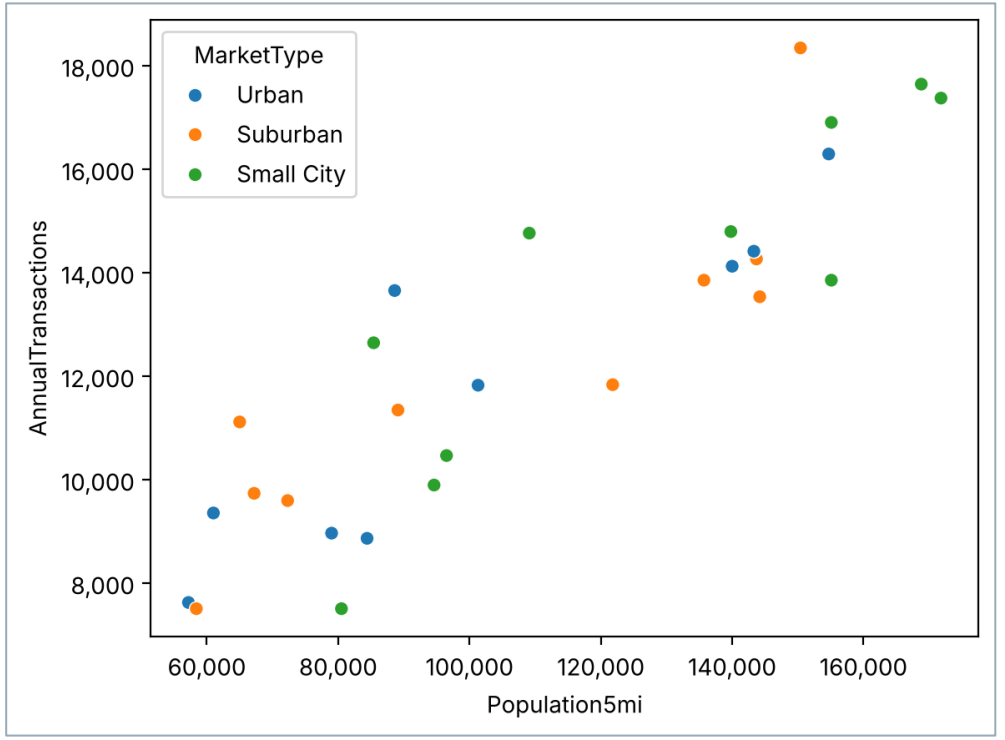
```
ax = sns.scatterplot(data=stores,
        x="Population5mi",
        y="AnnualTransactions",
        hue="MarketType")
ax.xaxis.set_major_formatter("{x:,.0f}")
ax.yaxis.set_major_formatter("{x:,.0f}")
```

Arguments

`x=` horizontal variable

`y=` vertical variable

`hue=` colors the points by a category



Regression Plot: Competition

Numerical · 2 variables

Flat line (r = -0.03). Nearby competitors show no association with transactions in this data.

Command

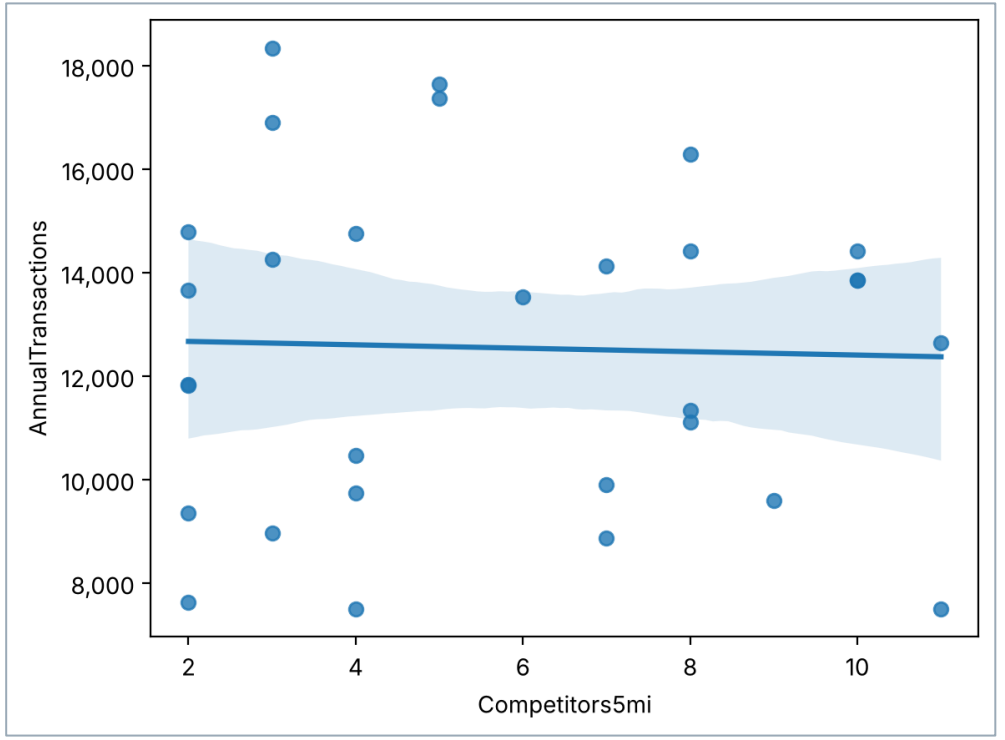
```
ax = sns.regplot(data=stores,
        x="Competitors5mi",
        y="AnnualTransactions")
ax.yaxis.set_major_formatter("{x:,.0f}")
```

Arguments

x= the possible predictor

y= the outcome

ci= band width, default 95 (None removes it)



Regression Plot: Income

Numerical · 2 variables

Slight downward slope (r = -0.37). Careful: lower-income areas here also tend to have more people.

Command

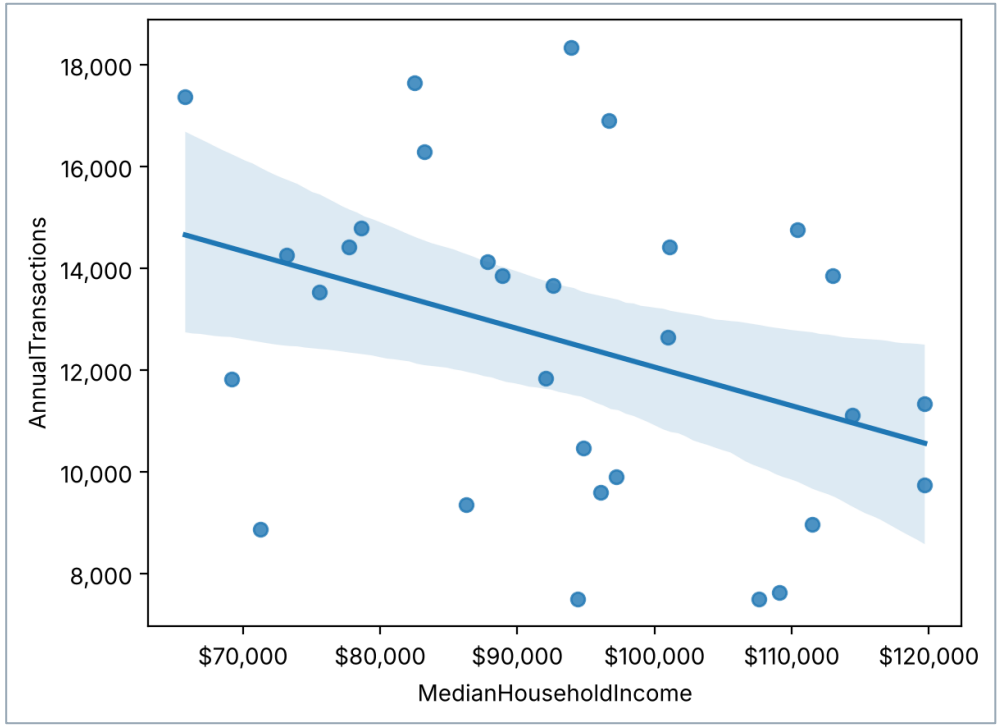
```
ax = sns.regplot(data=stores,
        x="MedianHouseholdIncome",
        y="AnnualTransactions")
ax.xaxis.set_major_formatter("${x:,.0f}")
ax.yaxis.set_major_formatter("{x:,.0f}")
```

Arguments

`x=` the possible predictor

`y=` the outcome

`"${x:,.0f}"` dollars with commas



Question 3

Which one predictor would you select for a simple regression model, and why?

Regression Plot: Population

Numerical · 2 variables

Our pick: r = 0.89, so population alone explains about 79% of the variation. Roughly +76 transactions per 1,000 people.

Command

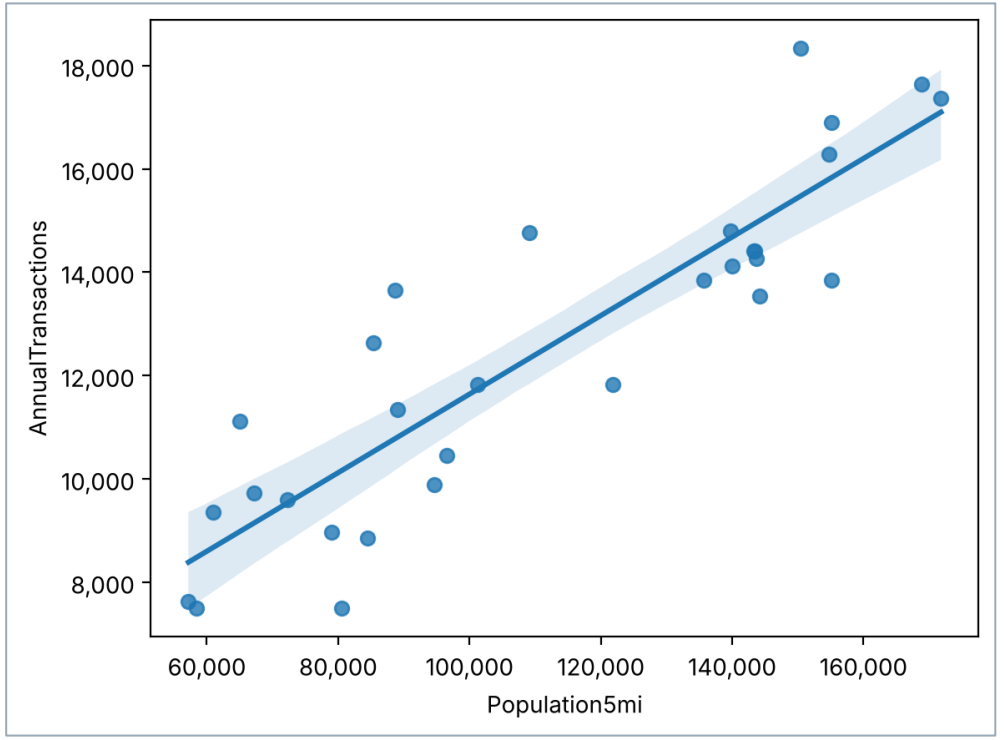
```
ax = sns.regplot(data=stores,
        x="Population5mi",
        y="AnnualTransactions")
ax.xaxis.set_major_formatter("{x:,.0f}")
ax.yaxis.set_major_formatter("{x:,.0f}")
```

Arguments

`x=` the selected predictor

`y=` the outcome

`ci=` 95% band around the line



Grouped Regression (lmplot)

Numerical · 2 variables, colored by 1 categorical

Separate lines for Standard and Compact stores. Both rise with population; Compact rises faster.

Command

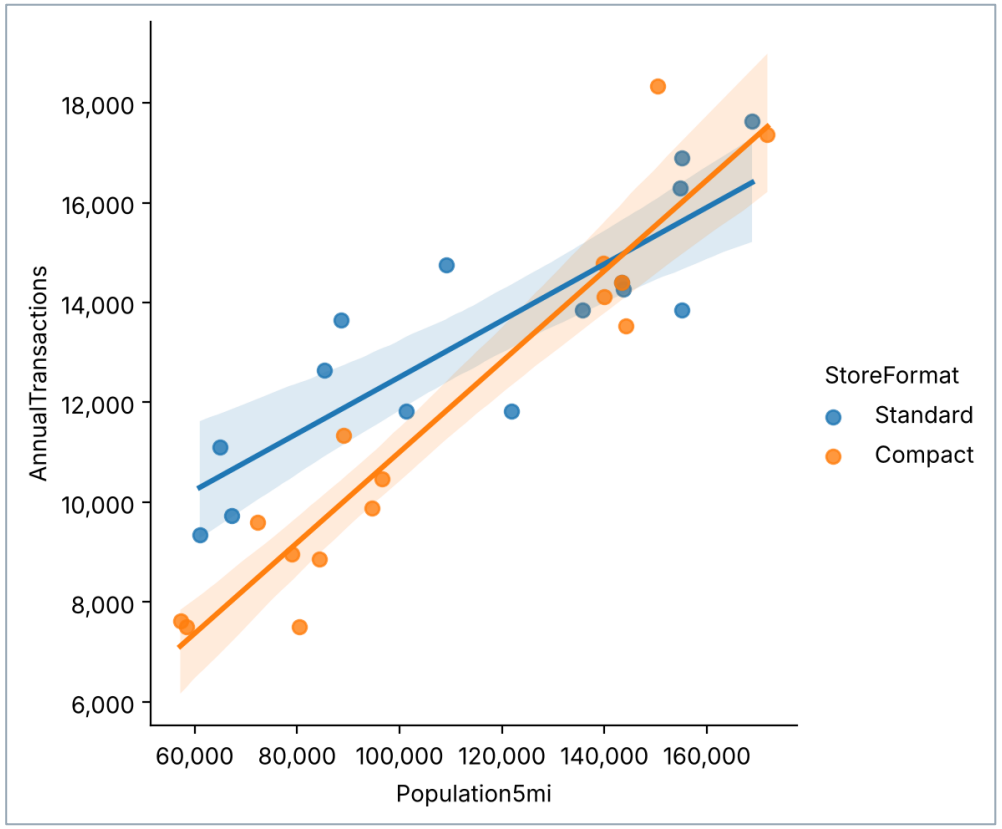
```
g = sns.lmplot(data=stores,
        x="Population5mi",
        y="AnnualTransactions",
        hue="StoreFormat")
for ax in g.axes.flat:
    ax.xaxis.set_major_formatter("{x:,.0f}")
    ax.yaxis.set_major_formatter("{x:,.0f}")
```

Arguments

`hue=` one line per category

`col=` one panel per category instead

`height=` panel size in inches



Question 4

What limitations or unusual observations should management consider when using the regression forecast?

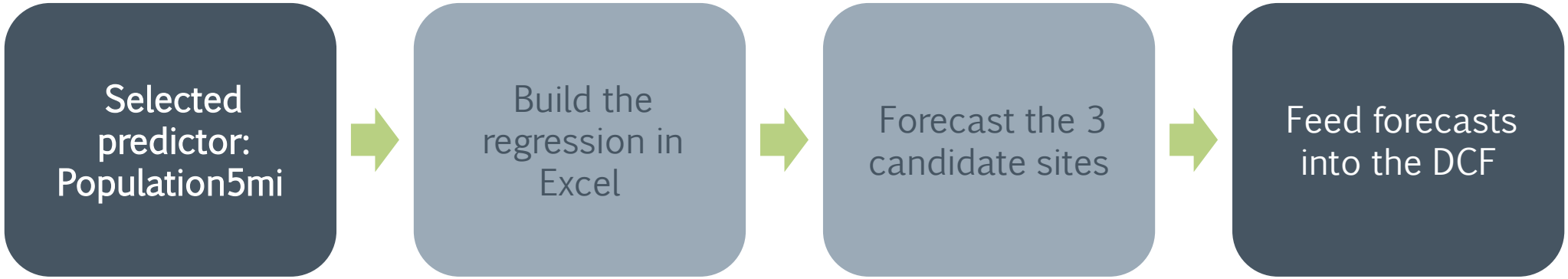
Limitations for Management

What the forecast can and cannot tell us

- Association, not causation: EDA shows what moves together, not why
- The simulation's ±10% transaction error is our assumption, not something the regression establishes
- Forecast only within the data: population 57,200 to 171,800
- Typical miss is about ±1,400 transactions; four stores missed by 2,400+
- 30 simulated stores: real data will be messier
- AnnualSales is calculated from transactions; never use it as a predictor

Next: Back to Excel

From predictor to decision



Python's job ends when we can name a single predictor and explain why.